Planning and Learning

Tutorial 13

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12.1 *Exam Question: Model based learning

The below questions appeared as sub-questions in various previous exams:

1. Using a learned transition model is mostly useful if only little data from the environment can be obtained. Give two reasons why it could be the case that only little data can be obtained, with an example environment to illustrate each reason. For example, your answer might consist of two sentences of the form "In environment X, the data that can be obtained is limited because of Y".

1. Possible reasons with corresponding environments are: data collection requires human supervision/intervention (e.g. in robotics); data collections is risky (e.g., industrial applications, autonomous driving); data collection requires human input (humans are part of the environment) (e.g., chatbot assistant). Relatedly: exploration can give a bad experience to users (e.g. recommendations in online shoppping); data collection is very computational expensive (e.g. detailed physics or chemistry simulators might take hours to compute a single time step); data collection has a monetary cost (eg. to pay people as above for chatbot assistant; cost for sub-optimal exploration actions as in a trading bot).

2. Consider the following statements:

   (a) Learning with a model becomes increasingly attractive when computational resources get more expensive.

   (b) Learning with a model becomes increasingly attractive when gathering data from the real system gets cheaper/faster.

- ☐ a) is false, but b) is true
- ☐ a) and b) are both false
- ☐ a) and b) are both true
- ☐ a) is true, but b) is false

2. a) and b) are both false.

3. Explain a similarity between Tabular Dyna-Q and the 'experience replay' technique used in Q-learning.

3. In both cases we do updates using experience tuples (s, a, r, s') from earlier time steps. (In case of replay these were stored in a batch, in case of Dyna-Q they were stored in the model)

12.2 *Exam Question: MCTS and AlphaGo

1. How does MCTS manage the exploration/exploitation trade off during the search?

1. By using an UCB criterium (which functions as an exploration bonus that gets lower with each visit).

2. AlphaGo can be seen as an adaption of Monte Carlo Tree Search which uses Neural Networks. Briefly explain the two functions of the Neural Network(s), compared to 'standard' MCTS.

2. (1) Evaluate / estimate the value / winning probability of a state / position (2) Define a prior / initial probability distribution over the actions / moves (to guide the search)

3. Why do we expect the tree search policy to be better than the 'raw' Neural Network policy?

3. The raw neural network policy is limited by the capacity of the neural network / due to function approximation error / task is hard for raw policy. The tree search policy can improve it performing lookahead search (which can focus on the current state).